HDOC Day-73

Enumeration of Days and Traffic Lights in C

Problem Statement

1. Create an enumeration for SUNDAY through SATURDAY and print each day with its integer value. 2. Define an enumeration for RED, YELLOW, GREEN and print Stop, Wait, or Go based on the selected light. A menu-driven program is used to demonstrate both operations.

Algorithm

1. Start the program and declare enum Day with SUNDAY through SATURDAY (default values 0 through 6).
2. Declare enum TrafficLight with RED = 0, YELLOW = 1, and GREEN = 2.
3. Display the menu: 1 Days, 2 Traffic light, 3 Exit. Read the choice.
4. For choice 1, iterate through the day enumeration and print the day names with their integer values.
5. For choice 2, read a traffic-light integer (0, 1, or 2). Use switch-case to print Stop, Wait, or Go. Reject invalid values.
6. For choice 3, exit. For any other menu choice, print an error. Repeat until Exit.

Test Case Table

TC	Menu choice	Input / Action	Expected result
1	1	No further input	SUNDAY = 0 through SATURDAY = 6
2	2	Light = 0 (RED)	Stop
3	2	Light = 1 (YELLOW)	Wait
4	2	Light = 2 (GREEN)	Go
5	2	Light = 3	Invalid traffic light value.
6	2	Light = -1	Invalid traffic light value.
7	9	Invalid menu choice	Invalid choice. Enter 1 to 3.
8	3	Exit	Exiting program.
9	1, then 3	Display days, then exit	Seven days and then Exiting program.
10	2, then 2, then 3	Light = 0, then Light = 2, then exit	Stop, Go, Exiting program.

C Program

```
#include <stdio.h>

enum Day { SUNDAY, MONDAY, TUESDAY, WEDNESDAY, THURSDAY, FRIDAY, SATURDAY };
enum TrafficLight { RED, YELLOW, GREEN };

void print_days(void)
{
    const char *names[] = {"SUNDAY", "MONDAY", "TUESDAY", "WEDNESDAY", "THURSDAY", "FRIDAY", "SATURDAY"};
    enum Day day;
```

```
    for (day = SUNDAY; day <= SATURDAY; day = (enum Day)(day + 1))
        printf("%s = %d\n", names[day], day);
}

void traffic_action(void)
{
    enum TrafficLight light;
    int choice;
    printf("Enter traffic light (0=RED, 1=YELLOW, 2=GREEN): ");
    if (scanf("%d", &choice) != 1) {
        printf("Invalid input.\n");
        return;
    }
    if (choice < RED || choice > GREEN) {
        printf("Invalid traffic light value.\n");
        return;
    }
    light = (enum TrafficLight)choice;
    switch (light) {
        case RED: printf("Stop\n"); break;
        case YELLOW: printf("Wait\n"); break;
        case GREEN: printf("Go\n"); break;
    }
}

int main(void)
{
    int choice;
    do {
        printf("\n--- HDOC DAY-73 MENU ---\n");
        printf("1. Display days and enumeration values\n");
        printf("2. Traffic light action\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        if (scanf("%d", &choice) != 1) {
            printf("Invalid input.\n");
            return 1;
        }
        switch (choice) {
            case 1: print_days(); break;
            case 2: traffic_action(); break;
            case 3: printf("Exiting program.\n"); break;
            default: printf("Invalid choice. Enter 1 to 3.\n");
        }
    } while (choice != 3);
    return 0;
}
```

Compilation and Execution Commands

gcc hdoc73.c -o hdoc73

./hdoc73

Screenshot Section

Compilation and Execution

```
┌──(kali㉿kali)-[~/Desktop]
└─$ vi hdoc73.c

┌──(kali㉿kali)-[~/Desktop]
└─$ gcc hdoc73.c -o hdoc73

┌──(kali㉿kali)-[~/Desktop]
└─$ ./hdoc73
```

Test Case 1

Menu choice: 1 | Input: No further input | Expected: SUNDAY = 0 through SATURDAY = 6

```
── HDOC DAY-73 MENU ──
1. Display days and enumeration values
2. Traffic light action
3. Exit
Enter your choice: 1
SUNDAY = 0
MONDAY = 1
TUESDAY = 2
WEDNESDAY = 3
THURSDAY = 4
FRIDAY = 5
SATURDAY = 6
```

Test Case 2

Menu choice: 2 | Input: Light = 0 (RED) | Expected: Stop

```
── HDOC DAY-73 MENU ──
1. Display days and enumeration values
2. Traffic light action
3. Exit
Enter your choice: 2
Enter traffic light (0=RED, 1=YELLOW, 2=GREEN): 0
Stop
```

Test Case 3

Menu choice: 2 | Input: Light = 1 (YELLOW) | Expected: Wait

```
── HDOC DAY-73 MENU ──
1. Display days and enumeration values
2. Traffic light action
3. Exit
Enter your choice: 2
Enter traffic light (0=RED, 1=YELLOW, 2=GREEN): 1
Wait
```

Test Case 4

Menu choice: 2 | Input: Light = 2 (GREEN) | Expected: Go

```
── HDOC DAY-73 MENU ──
1. Display days and enumeration values
2. Traffic light action
3. Exit
Enter your choice: 2
Enter traffic light (0=RED, 1=YELLOW, 2=GREEN): 2
Go
```

Test Case 5

Menu choice: 2 | Input: Light = 3 | Expected: Invalid traffic light value.

```
── HDOC DAY-73 MENU ──
1. Display days and enumeration values
2. Traffic light action
3. Exit
Enter your choice: 2
Enter traffic light (0=RED, 1=YELLOW, 2=GREEN): 3
Invalid traffic light value.
```

Test Case 6

Menu choice: 2 | Input: Light = -1 | Expected: Invalid traffic light value.

```
── HDOC DAY-73 MENU ──
1. Display days and enumeration values
2. Traffic light action
3. Exit
Enter your choice: 2
Enter traffic light (0=RED, 1=YELLOW, 2=GREEN): -1
Invalid traffic light value.
```

Test Case 7

Menu choice: 9 | Input: Invalid menu choice | Expected: Invalid choice. Enter 1 to 3.

```
── HDOC DAY-73 MENU ──
1. Display days and enumeration values
2. Traffic light action
3. Exit
Enter your choice: 9
Invalid choice. Enter 1 to 3.
```

Test Case 8

Menu choice: 3 | Input: Exit | Expected: Exiting program.

```
── HDOC DAY-73 MENU ──
1. Display days and enumeration values
2. Traffic light action
3. Exit
Enter your choice: 3
Exiting program.
```

Test Case 9

Menu choice: 1, then 3 | Input: Display days, then exit | Expected: Seven days and then Exiting program.

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./hdoc73

── HDOC DAY-73 MENU ──
1. Display days and enumeration values
2. Traffic light action
3. Exit
Enter your choice: 1
SUNDAY = 0
MONDAY = 1
TUESDAY = 2
WEDNESDAY = 3
THURSDAY = 4
FRIDAY = 5
SATURDAY = 6

── HDOC DAY-73 MENU ──
1. Display days and enumeration values
2. Traffic light action
3. Exit
Enter your choice: 3
Exiting program.
```

Test Case 10

Menu choice: 2, then 2, then 3 | Input: Light = 0, then Light = 2, then exit | Expected: Stop, Go, Exiting program.

